

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

DPR: History and Physical Exam Pediatrics Part 1: Birth to 5 years old

PPOM1

Ashka Patel, DO, FAAP

Associate Professor

Clinical Department of Pediatrics

Serota 123

Email: apate256@nyit.edu



Learning Objectives:

1. Basic Newborn Visit Interviewing/History Taking
2. Newborn Growth
3. Newborn Development
4. Vaccine Schedule
5. How to examine a newborn
6. Physical exam pathologies
7. Children – toddlers to 5 years of age history questions
8. Early AAP Screenings
9. Tips and Tricks to examine young children
10. Resources

At delivery -

A P G A R score (at 1 min and 5 mins of life)

At initial newborn visit -

Always ask birth history:

Gestational age

Birth weight if they have it

Complications during delivery?

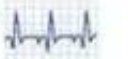
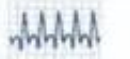
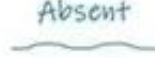
Did baby need NICU care/oxygen/phototherapy?

Mode of delivery? And why?

GBS status of mother or if mom had any diseases

Passed hearing, CCHD, got hep B, erythromycin, vit K? Blood type /coombs status

Target Oxygen Saturation Table	
2 minutes	65%-70%
3 minutes	70%-75%
4 minutes	75%-80%
5 minutes	80%-85%
10 minutes	85%-95%

The Apgar Score				
Indicator	0 Points	1 Points	2 Points	Total Points
A Appearance (skin color)	Blue (cyanotic) Pale (white) all over 	Pink body blue extremities (Acrocyanosis) 	Pink body 	—
P Pulse	Absent No pulse 	< 100 bpm 	> 100 bpm 	—
G Grimace Reflex Irritability	Floppy No response 	Minimal Response when Stimulation 	Prompt Response Sneezing, Crying or Pulling Away when stimulated 	—
A Activity (Muscle tone)	No Movement Flaccid 	Flexed arms and legs 	Active motion 	—
R Respiration	Absent No breathing 	Weak or Slow Irregular 	Vigorous Strong Cry 	—
Severely depressed Moderately depressed Excellent condition				0 - 3 4 - 6 7 - 10

Approach to the normal newborn

Weight: should be gaining 20-30g/day - should double weight by 4 months. Triple by 1 year
Normal FT Babies have 14 days to get back to their birth weight, want NO MORE THAN 10% weight loss

Asymmetric vs symmetric SGA, LGA

HC: Birth 35cm: growth fastest during newborn period - 2cm/month for the 3 months, then 1 cm/month for next 3 months - slows down to 0.5cm/month for 6 months. 1 year - fontanelles fuse

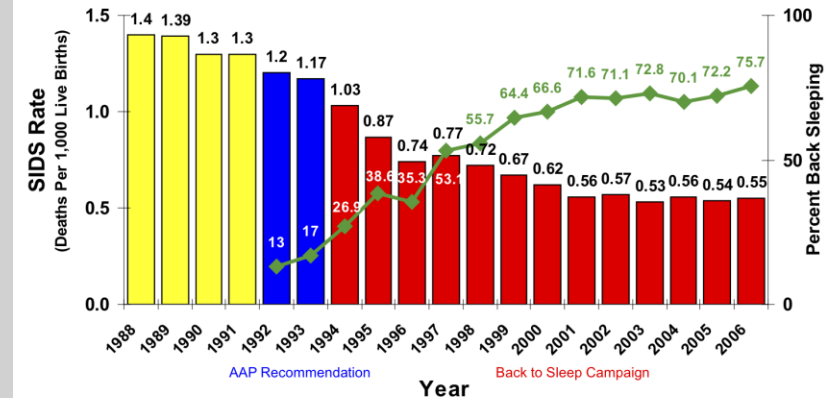
During this time: also important to check newborn screening results - if normal - can take up to 2 weeks to come back

If abnormal: usually clinic gets faxed important and emergent - might need redraw or intervention

Ask about feeding what and how much and how often, ask about elimination and urination, always ask about Safe Sleep and how mom and dad are doing. Use Edinburgh Postnatal depression scale

Happy mommy = Happy baby

SIDS Rate and Back Sleeping (1988 – 2006)



SIDS Rate Source: CDC, National Center for Health Statistics,
Sleep Position Data: NICHD, National Infant Sleep Position Study.



Approach to the normal newborn

Contraindications to Breast Feeding

Baby reasons: galactosemia, maple syrup urine disease, PKU

Parent reasons: HIV ½ (depending on treatment and viral load)

Active TB

HSV breast lesions

Ebola

Certain medications

Radioactive compounds

Breast feeding/formula exclusively until 4 months

4-6 months can start some purees/solids (one food at a time, early peanut exposure)

By age 1 can start cow's milk (max 24oz/day)

Stool colors you don't want: RED/WHITE/BLACK
all others are okay

Urine: urate crystals common

TABLE 2

Resources for Prescribing Medications During Lactation

For prescribers

LactMed

Free, peer reviewed, authoritative, and regularly updated e-Lactancia

Free, available in English and Spanish, peer reviewed, and regularly updated

Medications and Mothers' Milk

Authoritative textbook written by Thomas Hale, website and smartphone application require a subscription

For patients

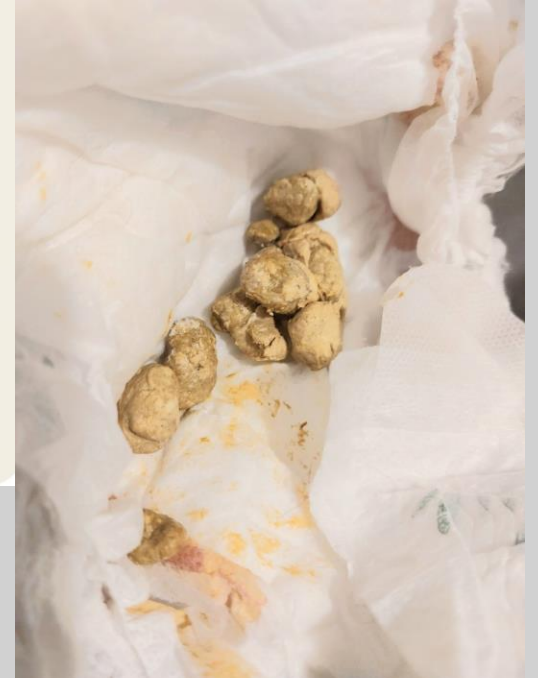
Drugs.com

Free general pharmaceutical database, includes well researched lactation information

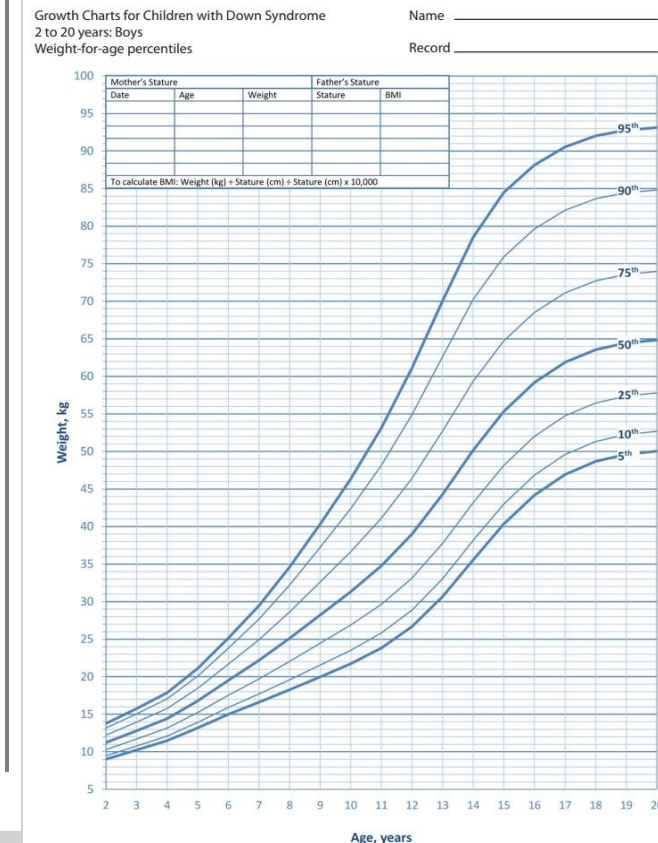
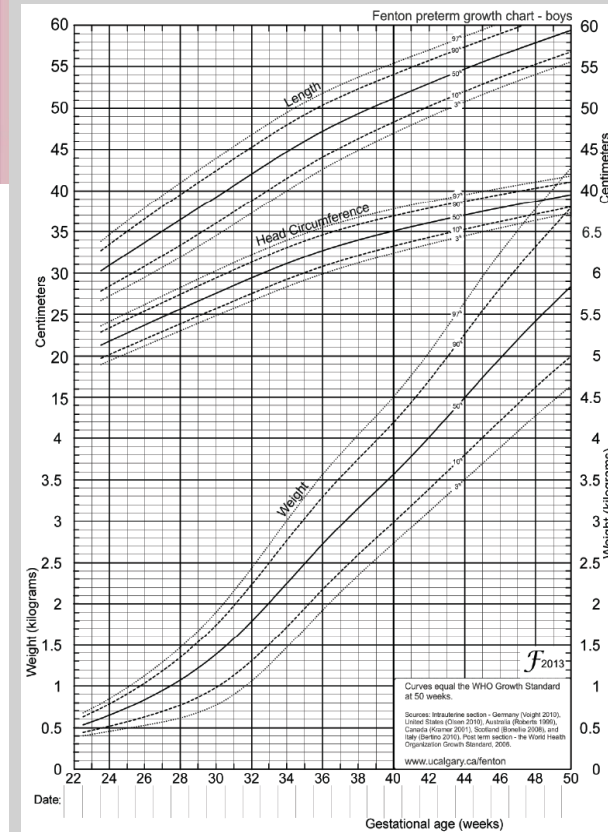
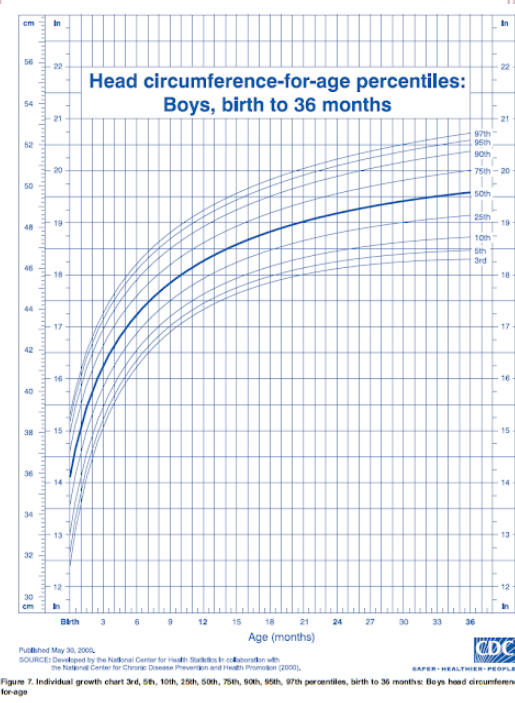
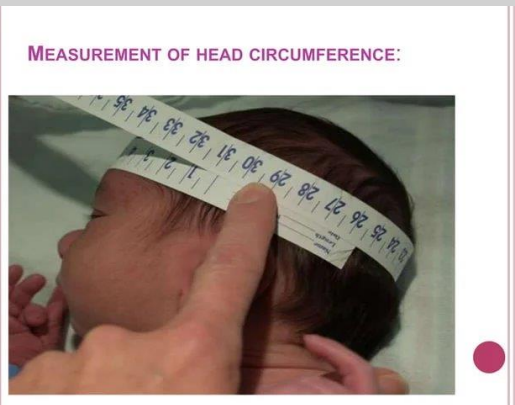
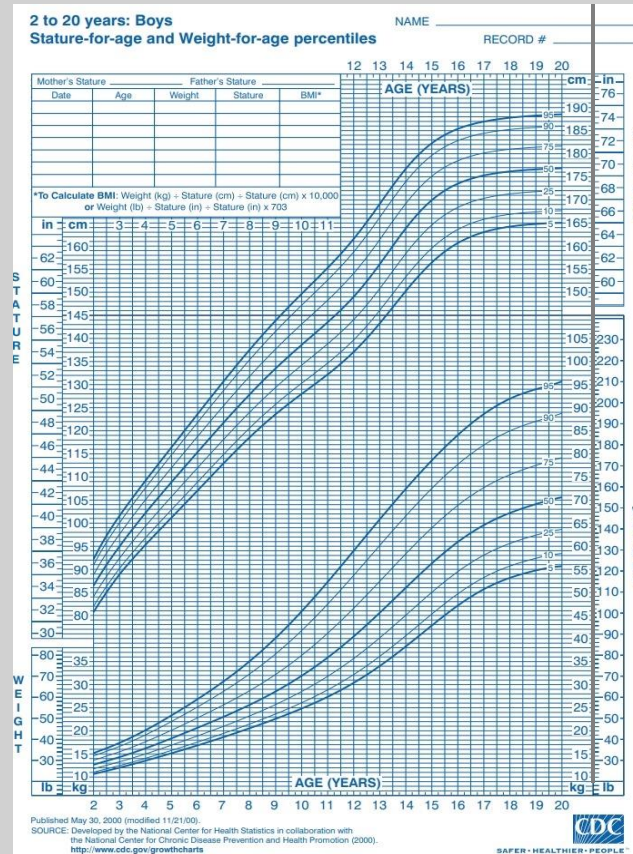
Mothertobaby.org

Free information on medications in pregnancy and lactation, available in English and Spanish

Information from references 2 and 4-7.



Growth charts



Development

Screening Scheme for Developmental Delay: Upper Range

AGE (MO)	GROSS MOTOR	FINE MOTOR	SOCIAL SKILLS	LANGUAGE
3	Supports weight on forearms	Opens hands spontaneously	Smiles appropriately	Coos, laughs
6	Sits momentarily	Transfers objects	Shows likes and dislikes	Babbles
9	Pulls to stand	Pincer grasp	Plays pat-a-cake, peek-a-boo	Imitates sounds
12	Walks with one hand held	Releases an object on command	Comes when called	One to two meaningful words
18	Walks upstairs with assistance	Feeds self from a spoon	Mimics actions of others	At least six words
24	Runs	Builds a tower of six blocks	Plays with others	Two- to three-word sentences

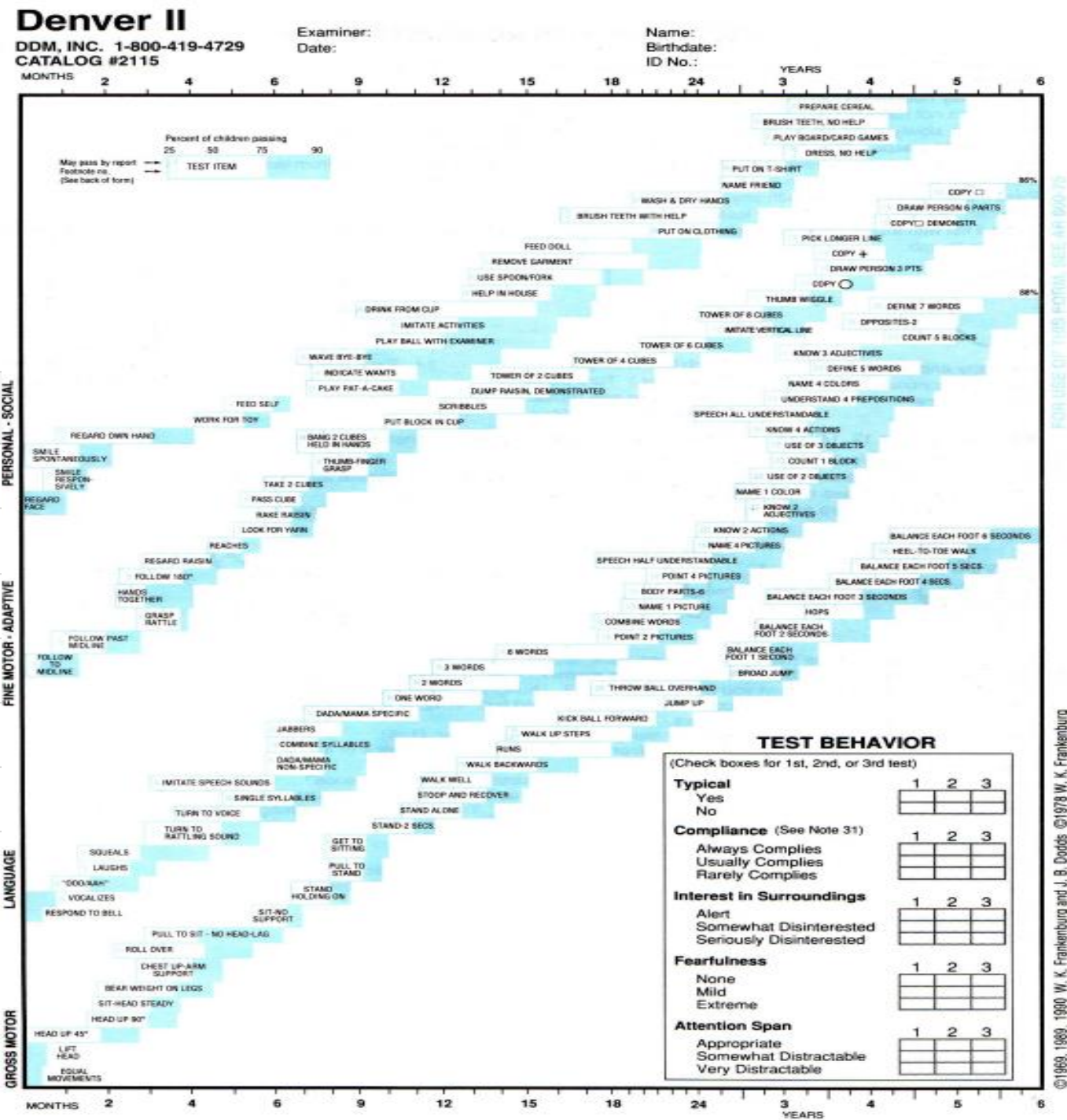


Table 1

Recommended Child and Adolescent Immunization Schedule
for Ages 18 Years or Younger, United States, 2025

American Academy of Pediatrics
DEDICATED TO THE HEALTH OF ALL CHILDREN®

These recommendations must be read with the Notes that follow. For those who fall behind or start late, provide catch-up vaccination at the earliest opportunity as indicated by the outlined purple bars. To determine minimum intervals between doses, see the catch-up schedule (Table 2).

Vaccine and other immunizing agents	Birth	1 mos	2 mos	4 mos	6 mos	8 mos	9 mos	12 mos	15 mos	18 mos	19–23 mos	2–3 yrs	4–6 yrs	7–10 yrs	11–12 yrs	13–15 yrs	16 yrs	17–18 yrs		
Respiratory syncytial virus (RSV-mAb [nirsevimab, clesrovimab])	1 dose during RSV season depending on maternal RSV vaccination status (See Notes)					1 dose nirsevimab during RSV season (See Notes)														
Hepatitis B (HepB)	1 st dose	2 nd dose			3 rd dose															
Rotavirus (RV): RV1 (2-dose series), RV5 (3-dose series)			1 st dose	2 nd dose	See Notes															
Diphtheria, tetanus, and acellular pertussis (DTaP <7 yrs)			1 st dose	2 nd dose	3 rd dose				4 th dose			5 th dose								
Haemophilus influenzae type b (Hib)			1 st dose	2 nd dose	See Notes			3 rd or 4 th dose (See Notes)												
Pneumococcal conjugate (PCV15, PCV20)			1 st dose	2 nd dose	3 rd dose			4 th dose												
Inactivated poliovirus (IPV)			1 st dose	2 nd dose	3 rd dose							4 th dose	See Notes							
COVID-19 (1vCOV-mRNA, 1vCOV-aPS)					1 or more doses of 2025–2026 vaccine (See Notes)							1 dose of 2025–2026 vaccine (See Notes)					1 dose of 2025–2026 vaccine (See Notes)			
Influenza					1 or 2 doses annually (See Notes)										1 dose annually (See Notes)					
Measles, mumps, and rubella (MMR)					See Notes			1 st dose				2 nd dose								
Varicella (VAR)								1 st dose				2 nd dose								
Hepatitis A (HepA)					See Notes			2-dose series (See Notes)												
Tetanus, diphtheria, and acellular pertussis (Tdap ≥7 yrs)															1 st dose					
Human papillomavirus (HPV)															2-dose series	See Notes				
Meningococcal (MenACWY-CRM ≥2 mos, MenACWY-TT ≥2years)			See Notes														1 st dose		2 nd dose	
Meningococcal B (MenB-4C, MenB-FHbp)															See Notes					
Respiratory syncytial virus vaccine (RSV [Abrysvo])															Seasonal administration during pregnancy if not previously vaccinated					
Dengue (DEN4CYD: 9–16 yrs)															Seropositive in areas with endemic dengue (See Notes)					
Mpox																				

Range of recommended ages for all children

Range of recommended ages for catch-up vaccination

Range of recommended ages for certain high-risk groups or populations

Recommended vaccination for those who desire protection

Recommended vaccination based on shared clinical decision-making

Newborn Exam

Start with Head - fontanelles, shape of head

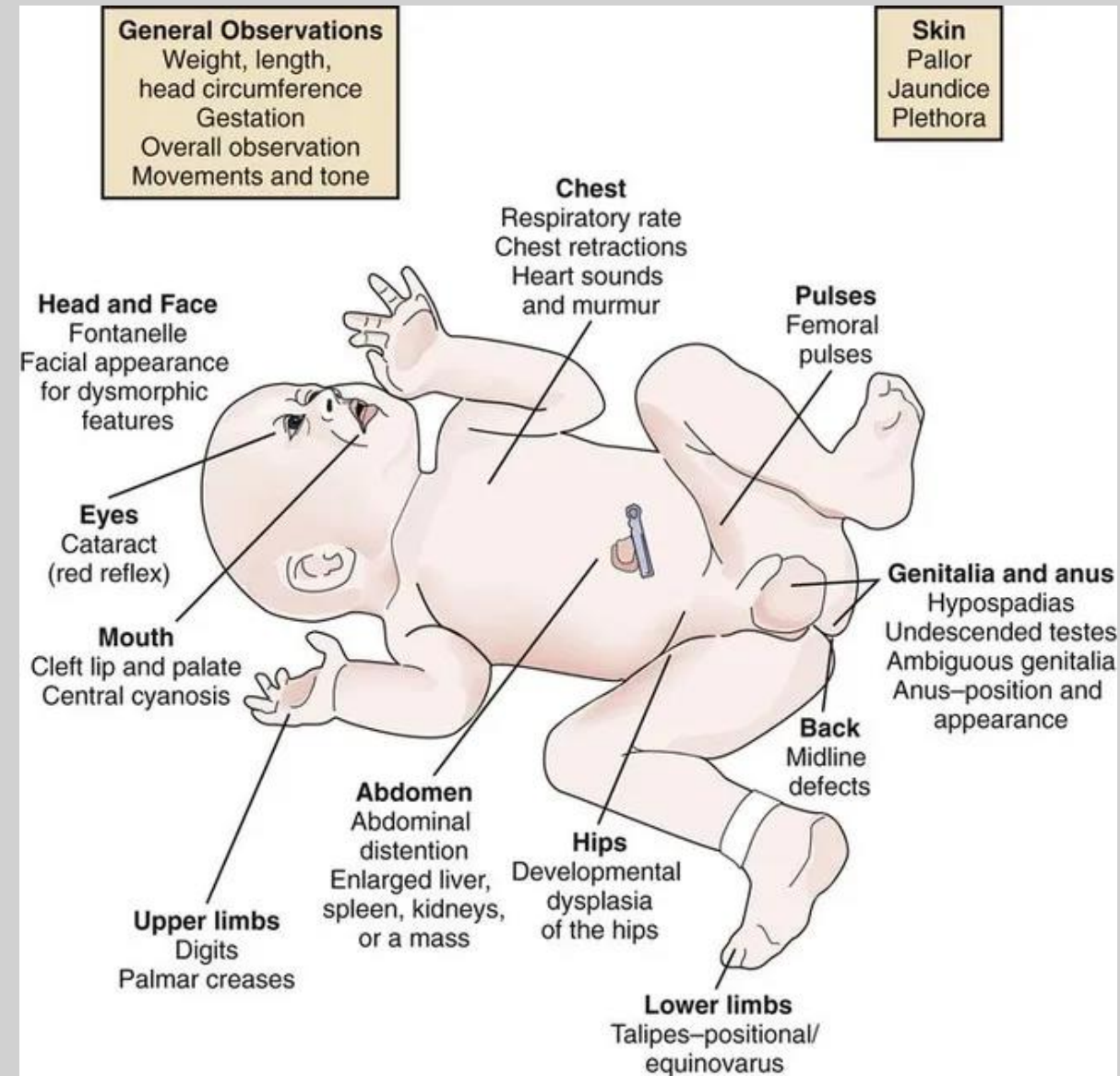
Eyes: RR reflex, spacing of eyes, strabismus, tracking by 2-4 months

Ears: pits, tags, low set vs not

Nose: Patent nares

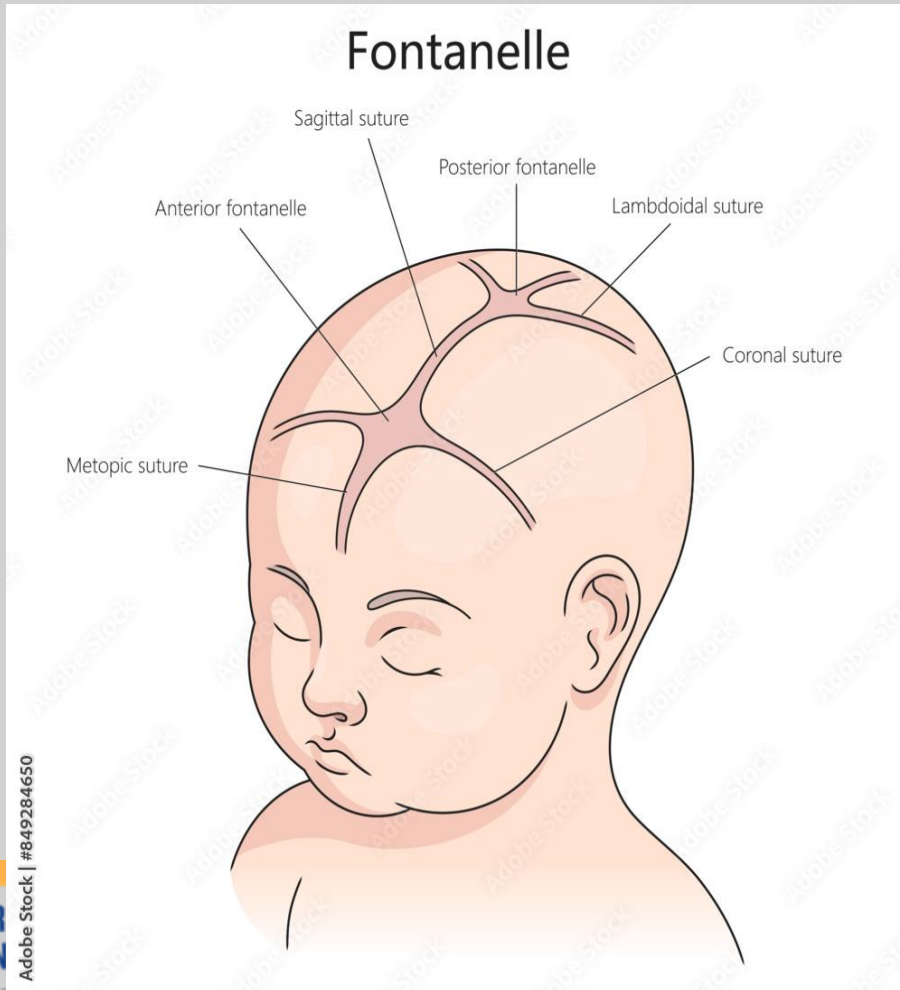
Mouth: outer Deformities, check suck reflex and hard palate intact

Clavicles (newborn): for deformities/step offs

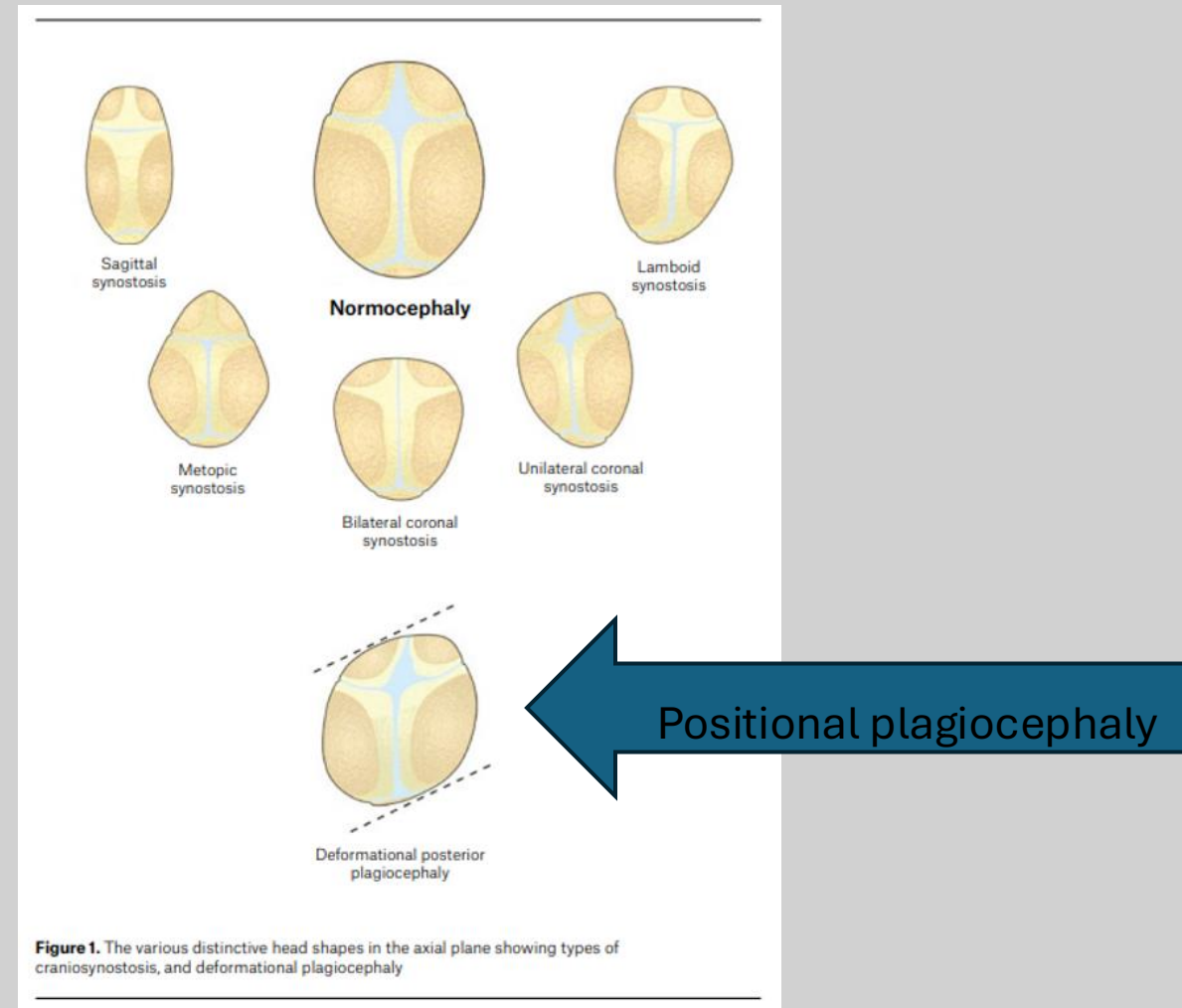


Newborn Exam

Normal:



Abnormal:



Types of scalp findings from delivery in newborn

Caput succedaneum – like a CAP swelling is throughout, crosses suture lines, usually prolonged delivery – due to birth canal or vacuum assisted. Resolves 24-48 hours

Cephalohematoma – DOES not cross suture lines, often with bruising, skin discoloration, Resolves 2-4 weeks, +can cause jaundice

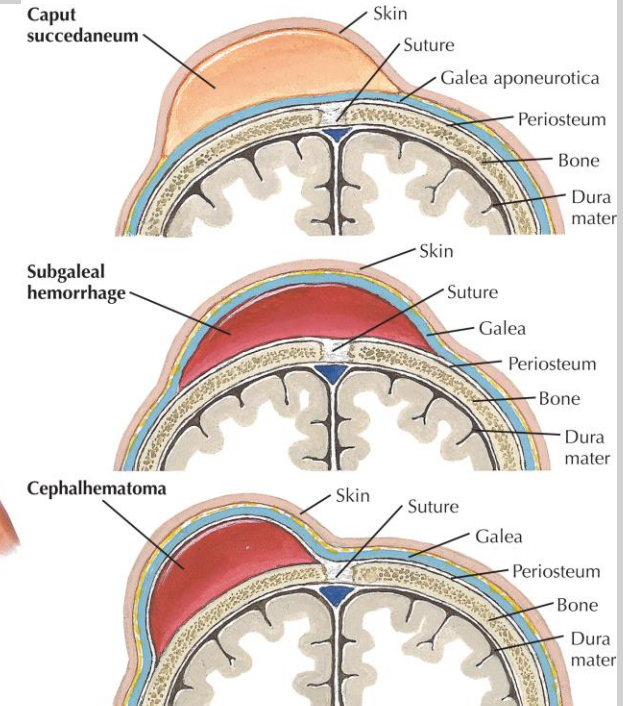
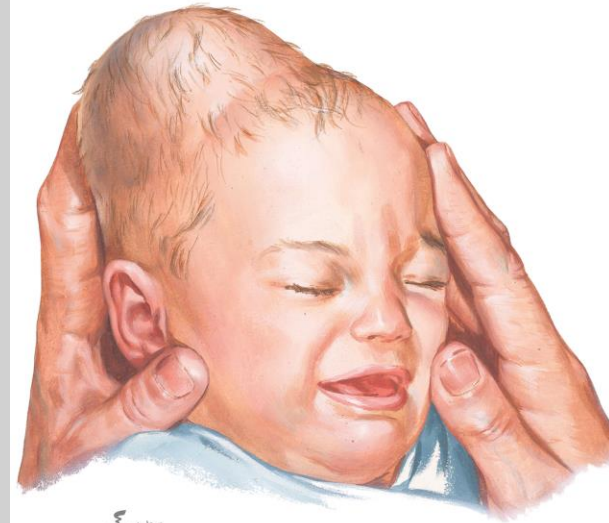
Subgaleal hemorrhage: dangerous, crosses sutures, Can accumulate a lot of blood causing anemia, Hypovolemia, +positive fluid wave

Caput succedaneum V/S Cephalohematoma



Dr. Rekha Thaddanee

Extracranial Hemorrhage or Edema in Newborn



Newborn Exam

Normal Red Reflex:



Abnormal Red Reflex:



Etiologies: congenital cataracts, retinoblastoma, chorioretinal coloboma, scarring from retinopathy of prematurity

TABLE 99.1

Visual System Milestones
From Edward DP, Kaufman LM. Anatomy, development, and physiology of the visual system. 2003;50:1–23.

Description	Age
Pupillary light reaction present	30 weeks' gestation
Pupillary light reaction well developed	1 month
Lid closure in response to bright light	30 weeks' gestation
Blink response to visual threat	2–5 months
Visual fixation present	Birth
Fixation well developed	2 months
Visual following well developed	3 months
Accommodation well developed	4 months
Visual evoked potential acuity at adult level	6 months
Grating acuity preferential looking at adult level	2 years
Snellen letter acuity at adult level	2 years
Color vision present	2 months
Color vision at adult level	6 months
Stereopsis developed	6 months
Stereoacuity at adult level	7 years
End of critical period for monocular visual deprivation	10 years

Newborn Exam

Neonatal conjunctivitis:

Incidence 1-21% depending on region

Gonococcal Infections: Copious purulent discharge (EARLY) can start 2-4 days of birth. Tx: IM dose of 3rd generation cephalosporin

Chlamydia Infections: Watery discharge (LATER) will start 5-14 days from birth. Tx: oral azithromycin (or erythromycin), not preventable with erythromycin ointment at birth

HSV: can cause infectious keratitis, scarring of the cornea

Swabs helpful!

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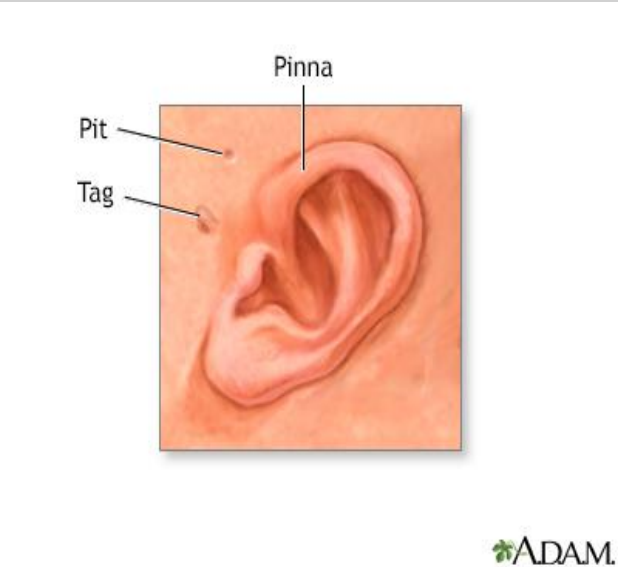
FIG. 20.57

Ophthalmia neonatorum, a hyperacute bacterial conjunctivitis, with thick purulent discharge and red swollen lids.

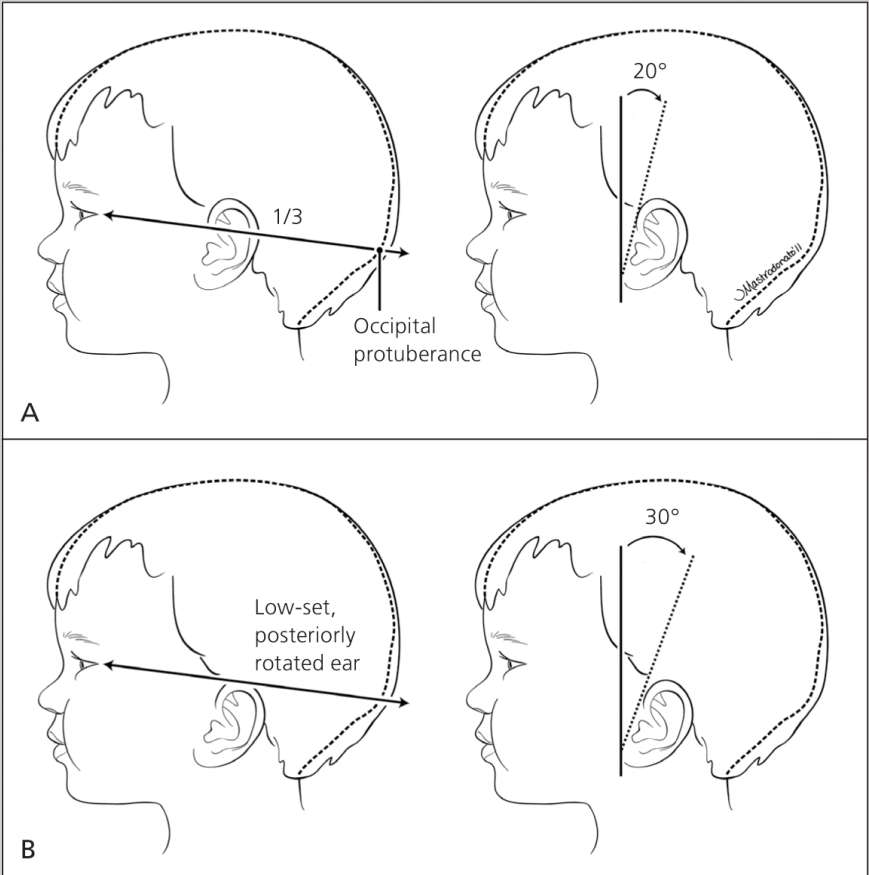


Newborn Exam

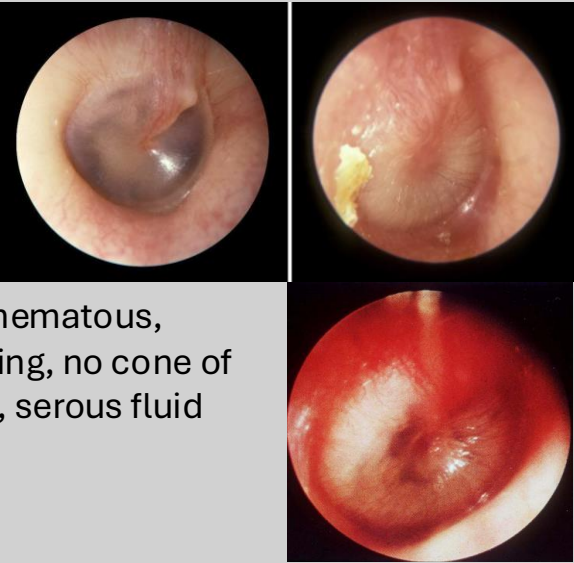
Ears



Positioning relative to head and lateral epicanthal folds

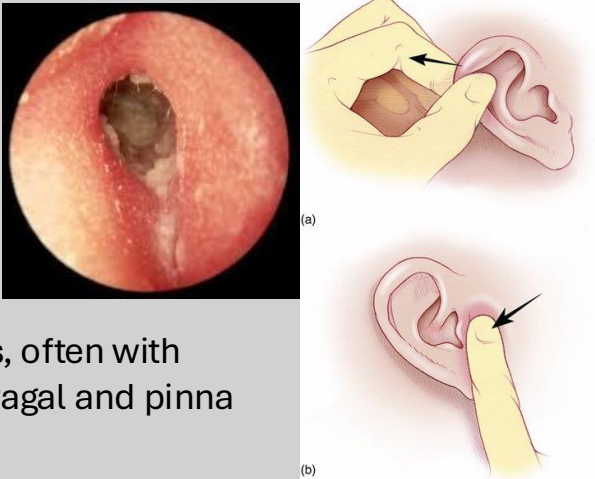


Otitis media



Erythematous, bulging, no cone of light, serous fluid

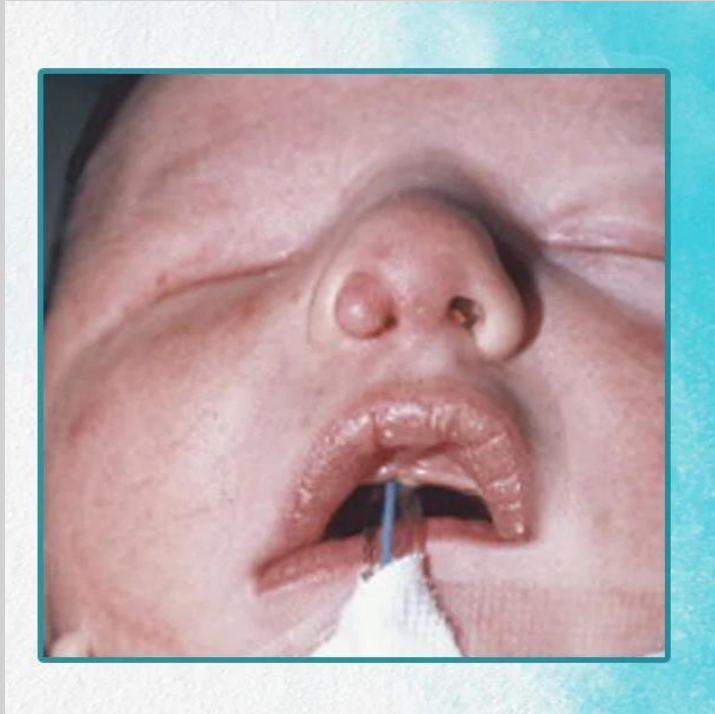
Otitis Externa



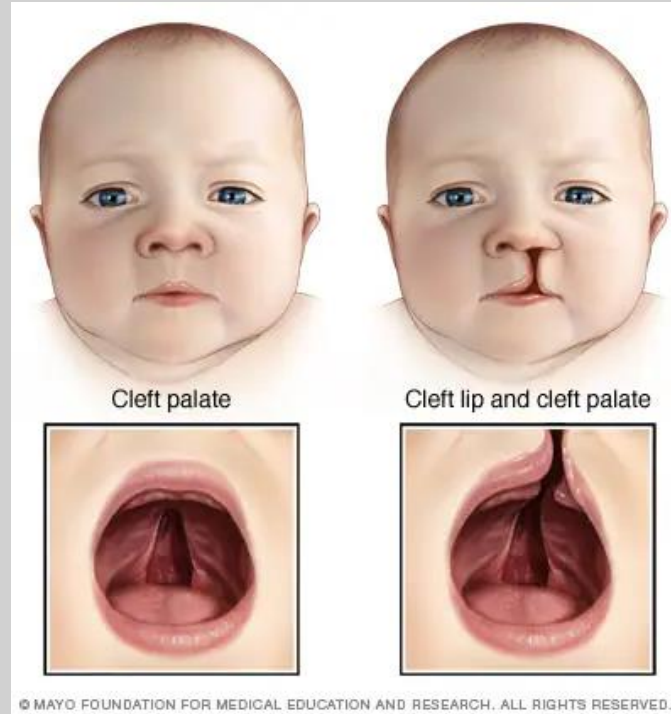
Canal erythematous, often with discharge, pain on tragal and pinna pulling

Newborn Exam

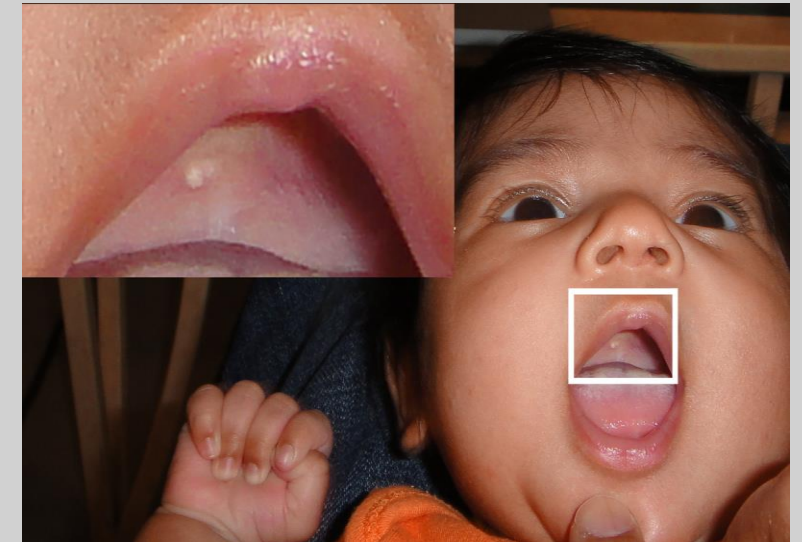
Nose and Mouth



Baby's are obligate nose breathers
Clonal atresia – unilateral or bilateral



Cleft palate vs Cleft lip



Epstein pearl (normal finding) can be seen on gums as well

Newborn Exam

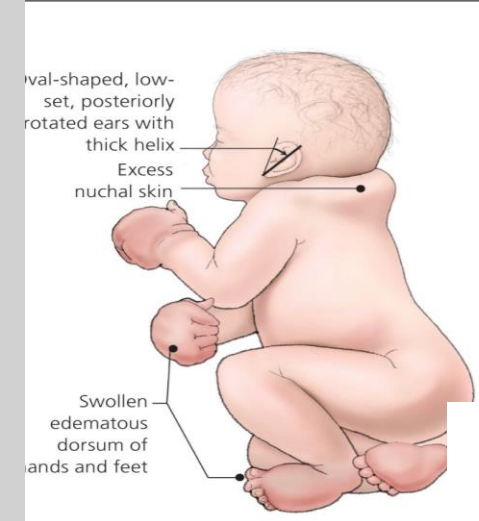
Clavicles



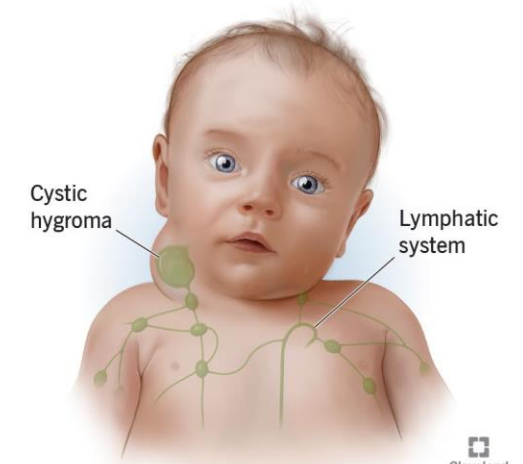
Clavicular Fx

Neck

Can be webbed, short, excess skin, cystic hygroma, cysts, torticollis



Cystic Hygroma



Newborn Exam

Upper Extremities: tone, strength, 5 fingers on each hand, creases, grasp, moro (disappears by about 2-4 months)

Chest: deformities, heart and lung sounds

Abdomen: distension, soft, masses, HSM, hernias

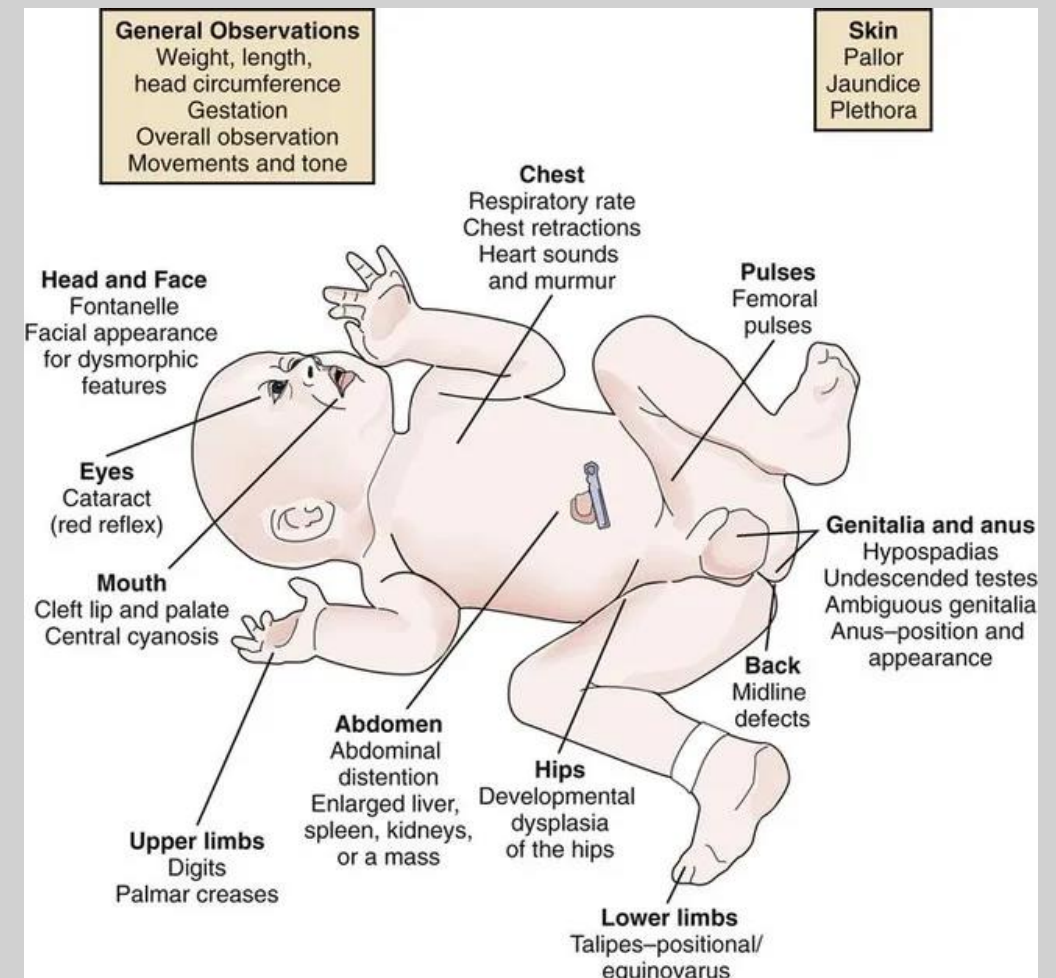
GU: femoral pulses bilaterally, ambiguous genitalia, males descended testes, anus patent!

Lower Extremities: Hips - Ortolani & Bartlow, leg length, tone, strength, deformities, 5 toes, plantar reflex

Back: Spine deformities, tufts of hair, sacral dimples

Neuro: reflexes

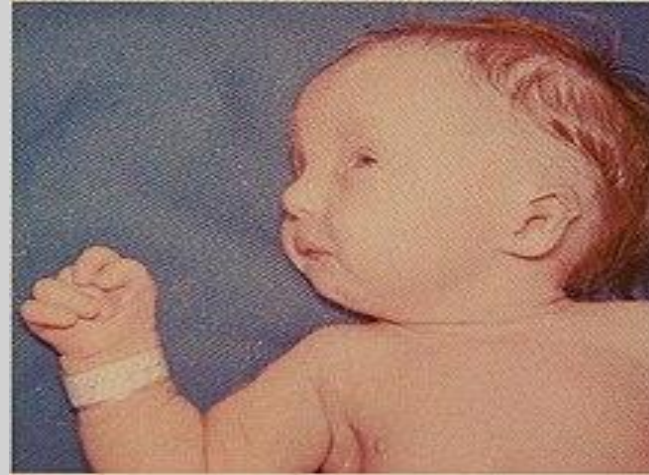
Skin: rashes, jaundice, etc



Newborn Exam

Hands

Overlapping clenched hands = Trisomy 13+18



Polydactyl



Feet

Syndactyly



Club feet – bilateral equinovarus deformities



Newborn Exam

Prominent xyphoid



Chest

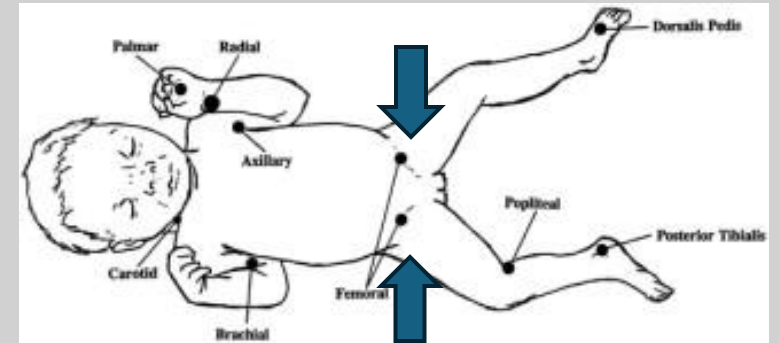
Pectus excavatum



Supernumerary nipple

Pulses

120-160



Femoral pulses very important! if abnormal = coarctation of aorta

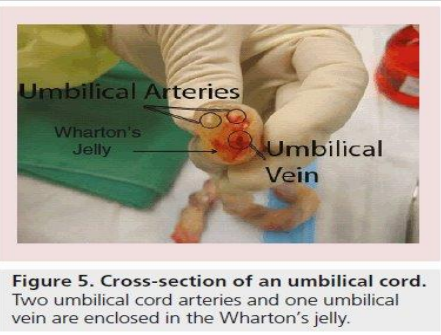
Abdomen



Check for hepatosplenomegaly and abdominal masses/hernias



Umbilical cord care



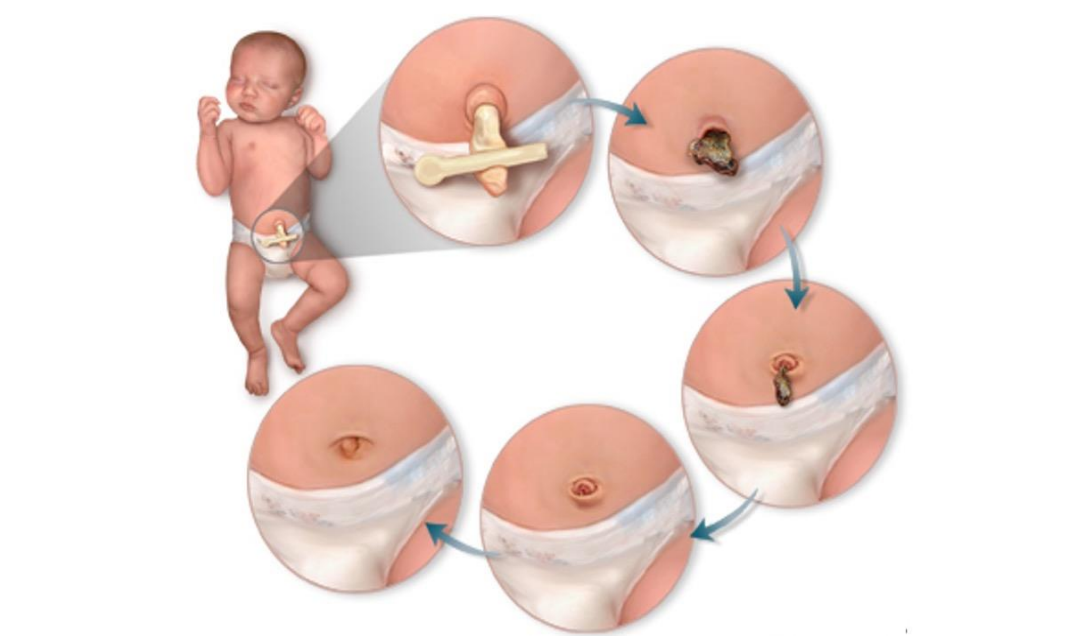
Falls off on it's own in 10-14 days – cannot submerge in water til then

Delayed separation: immune disorder

DO NOT cover

If bleeding or foul smelling discharge, alert PCP

Umbilical hernia – nothing to do until age 3-6 years, as long as it's compressible



[Enlarge](#) [Print](#)

Selected Differential Diagnosis of Neonatal Umbilical Masses

Condition	Characteristics
Umbilical pyogenic granuloma	Small, red, moist, velvety mass
Umbilical polyp	Firm, red mass; does not resolve after application of silver nitrate
Umbilical hernia	Covered with skin, usually reducible
Patent urachus	Discharge (urine) from the umbilicus
Omphalomesenteric duct remnant	Bilious or fecal discharge from umbilicus
Omphalocele	Membranous sac containing viscera herniates through the umbilicus
Gastroschisis	Loops of bowel protrude lateral to an intact umbilicus

Umbilical granuloma



Umbilical hernia

Newborn Exam

Heart:

Fast Rate

Murmurs hard to hear when subtle

Grading (1-6)

Systolic vs Diastolic

Better with positioning?

Cyanosis

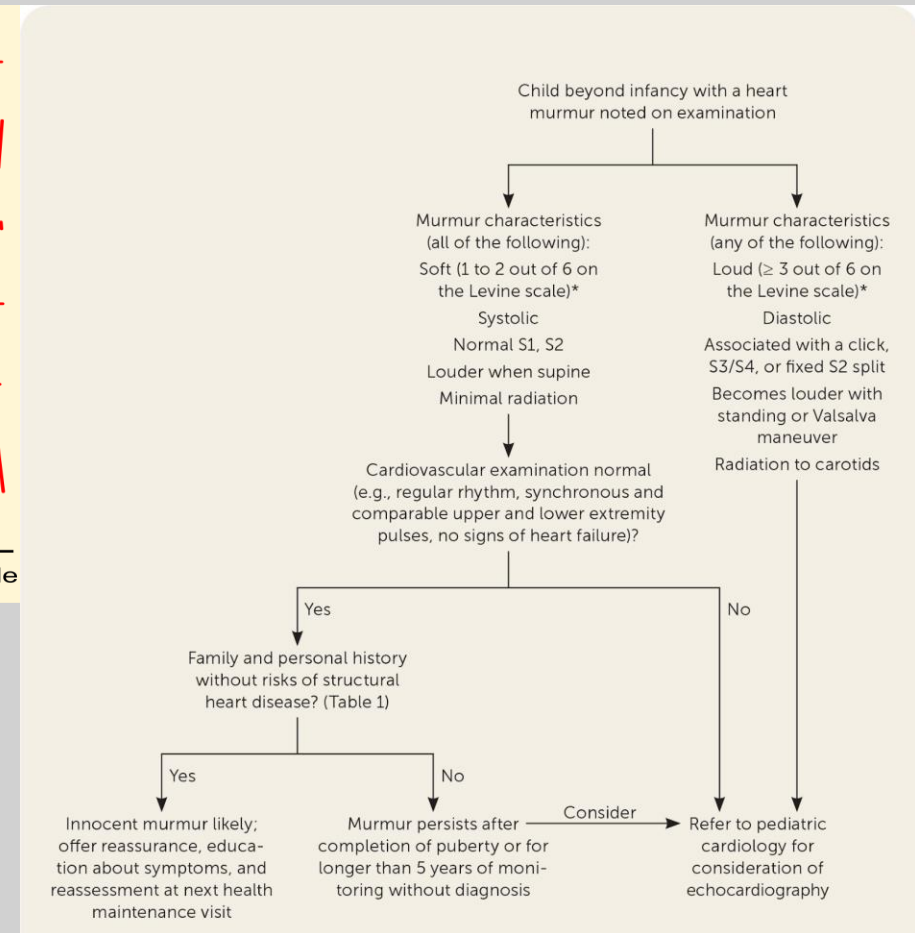
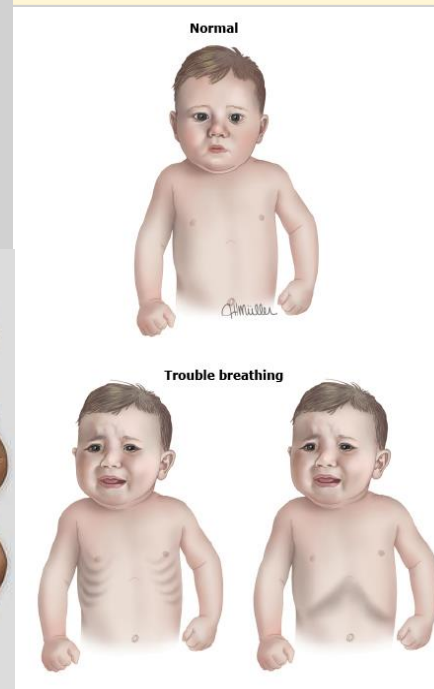
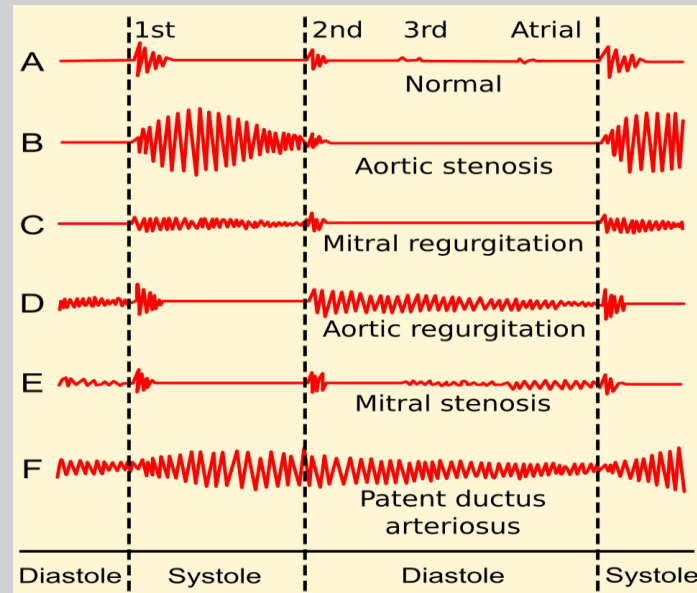
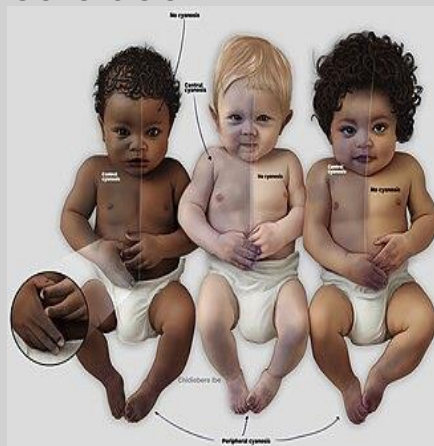
Lungs:

Lung sounds

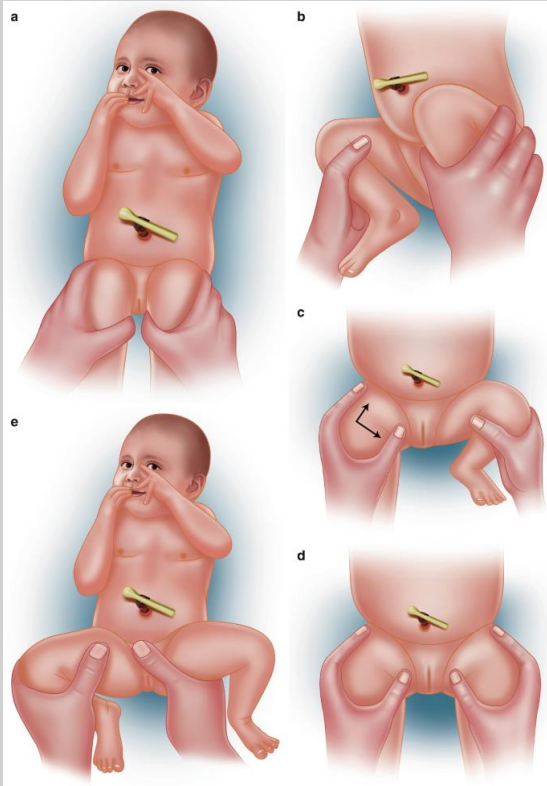
RR 30-60 normal in newborns

Signs of respiratory distress

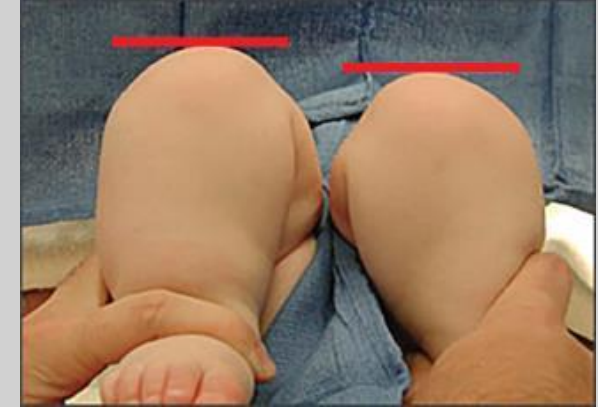
Accessory muscle use



Hip Exam: checking for Hip Dysplasia



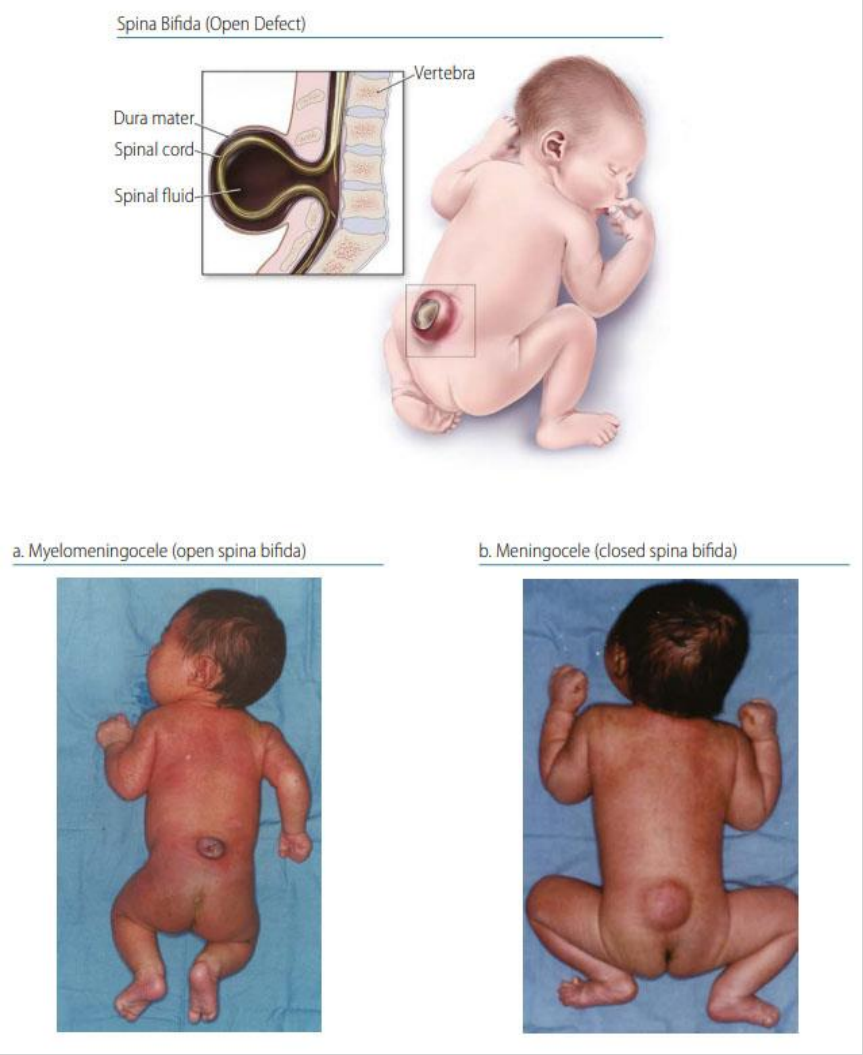
- Risk Factors: Female, Breech, Pos Family history
- **Barlow sign "Back"**
- Maneuver attempts to subluxate or dislocate unstable hip from acetabulum with gentle hip **adduction**
- **Ortolani sign "Out"**
- Maneuver attempts to reduce a subluxated or dislocated femoral head back into acetabulum with gentle hip **abduction**
- Asymmetry
- Asymmetrical gluteal or thigh skinfolds
- Specificity is low; finding is present in up to 25% of unaffected infants
- Younger than 4 months – b/I Hip US
- After 4 months – do bilateral hip xray



Positive Galeazzi Sign ↑



Back: Spina Bifida



Sacral dimple vs cleft

GU exam

Females:

Labia – make sure not fused

Clitoromegaly

Some white vaginal discharge + blood - normal – mom's hormones

Males:

Testicular exam – both descended ** **Cryptorchidism**

Urethra placement – hypospadias vs epispadias

Scrotal sac – make sure not bifid

Hydroceles give you positive transillumination test

In BOTH: Anus patent

If Ambiguous Genitalia in a Newborn : have to rule out classic **Congenital Adrenal Hyperplasia (CAH)

When to suspect

- Micropenis: Stretched penile length < 2.5cm in a term newborn
- Asymmetry of labioscrotal folds
- A penis with B/L nonpalpable testis
- U/L cryptorchidism with hypospadias
- Perineoscrotal or penoscrotal hypospadias with or without microphallus even if testes are descended.
- Apparently female external genitalia with clitoromegaly or inguinal hernia
- Overtly abnormal genitals like cloacal exstrophy
- Discordance of external genitalia with prenatal karyotype.



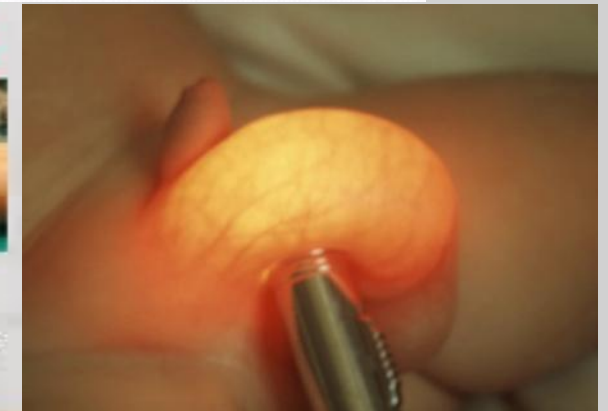
Imperforated Anus

If a baby is born without an anus (the opening at the end of the digestive tract), then this birth defect is known as an imperforated anus.

Baby is not able to pass stools from the body.

It is an **acute emergency**.

Surgery is required to repair imperforated anus.
Initial surgery to reroute the intestines (colostomy)
Second surgery for colostomy closure.

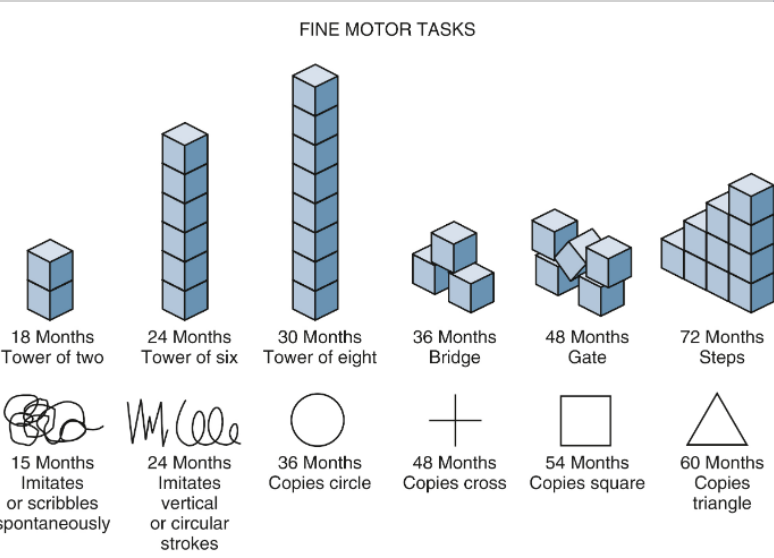
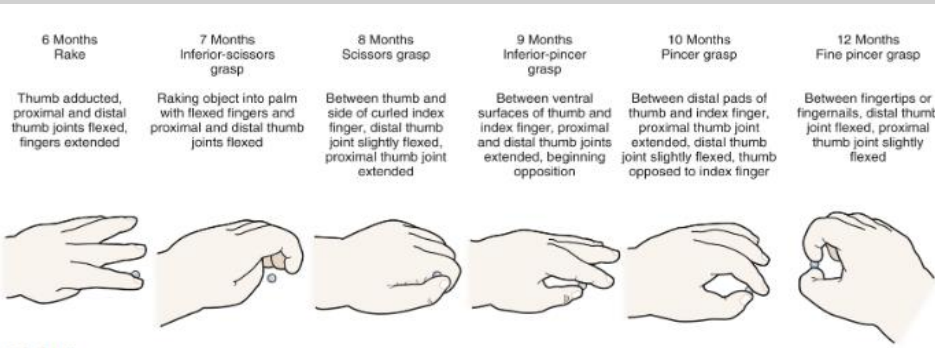


Neurologic/Development

Check out Tone
Hyper and hypo can be bad

Primitive Reflexes

Regression is BAD!



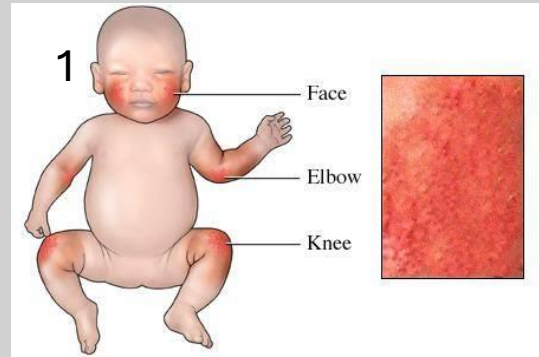
Primitive Reflexes and Protective Equilibrium Responses

Reflex	Appearance ^a	Disappearance ^a
Moro	Birth	4 months old
Hand grasp	Birth	3 months old
Crossed adductor	Birth	7 months old
Toe grasp	Birth	8–15 months old
ATNR	2 weeks old	6 months old
Head righting	4–6 months old	Persists voluntarily
Protective equilibrium	4–6 months old	Persists voluntarily
Parachute	8–9 months old	Persists voluntarily

ATNR, Asymmetric tonic neck reflex.

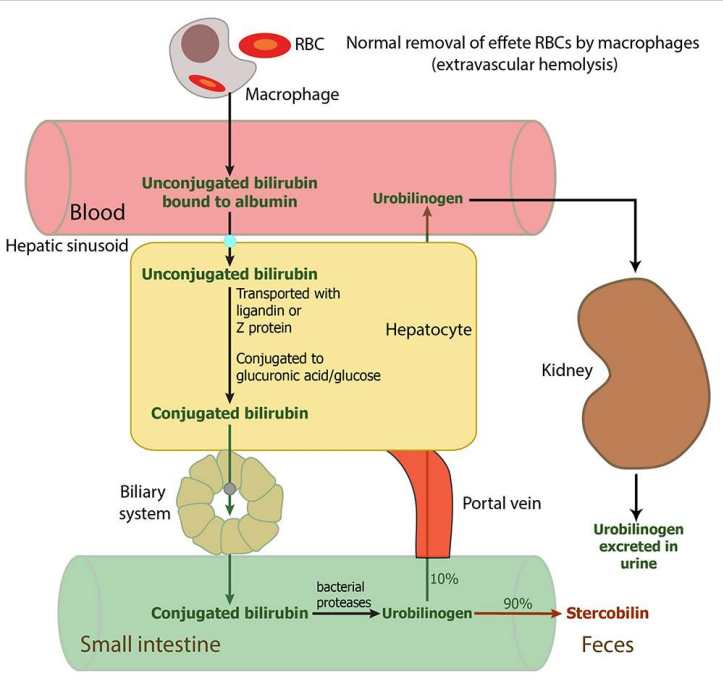
Skin:

1. Infantile Eczema
2. Hemangioma
3. Cradle Cap (seborrheic dermatitis)
4. Nevus Simplex
5. Erythema toxicum neonatorum (ETN) (Etox)
6. Pustular melanosis
7. Congenital Dermal Melanocytosis
8. Milia

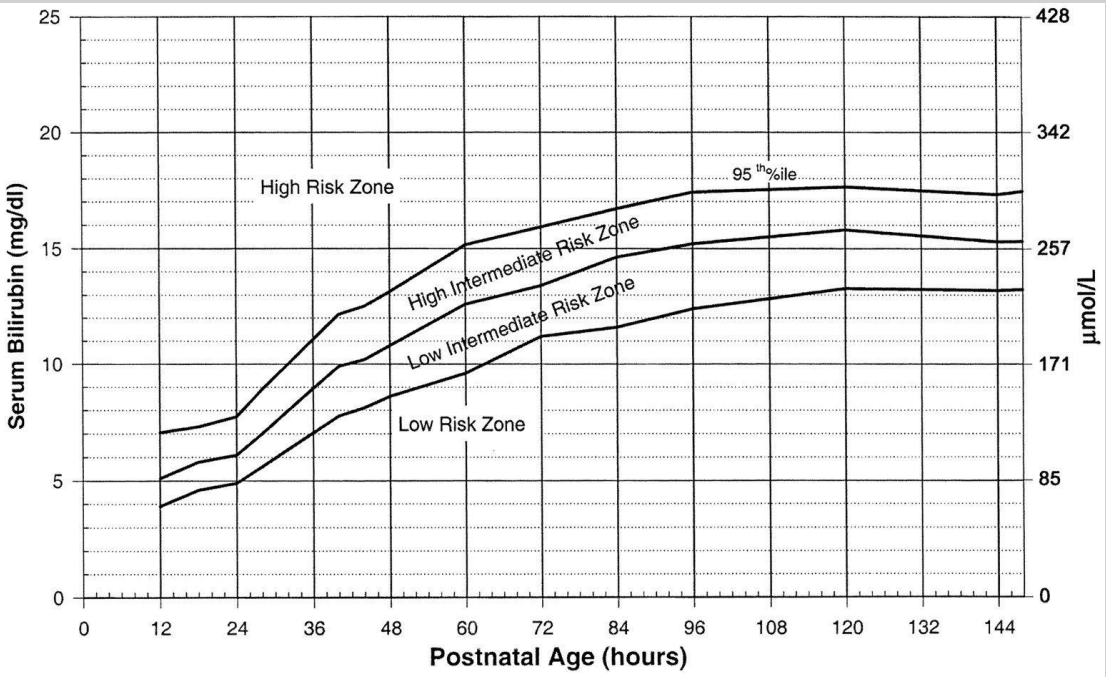


Phototherapy

Jaundice



Unconjugated Bilirubin
build up



Children - toddlers up to 5

Ask milestone questions: Gross motor, Fine motor, Language, Social Emotional

Screenings: **M-CHAT scoring** at 18 months, 24 months
Developmental screening: 9months, 18 months, 30 months (Ages and Stages Ques) vs surveillance at each well child visit

How is school/daycare going? Changes in social situation?

Safety questioning: guns, smoking, seat belts, car seats/booster seats, helmets while riding bike, smoke detectors, carbon monoxide detectors, poison control info at hand for toddlers

Anticipatory guidance: baby proofing house, locking away meds, cabinet + oven clamps

Hearing and Vision screenings in clinic – older than 3-4 years of age

AAP recommends **Lead and Iron testing at 1 yr plus/minus 2 yr** (pre-mie/WIC 9 months), BP measurement starting at age 3

Toilet Training: usually >18 months when child is ready



Do you know the four stages of car seat safety?



Rear Facing

Rear facing provides the best protection for a young child's spine. Rear face for a minimum of 2 years, ideally until the seat is outgrown.



Forward Facing

When a child has outgrown their rear facing convertible seat and is at least 2 years old, move to a forward facing 5 point harness.



Booster

When a child has outgrown their harnessed seat, is at least 5 years old, and is mature enough to sit properly, move to a belt positioning booster.



Adult Seat Belt

When a child is at least 10 years old and can pass the 5 step test, move to the seat belt alone.

Children - toddlers up to 5

TABLE 1 DSM-5 Criteria for Autism Spectrum Disorder

Domains	Criteria: Deficits	Examples
A. Persistent deficits in social communication and social interaction across multiple contexts, as manifested by the following, currently or by history; must have all 3 symptoms in this domain	1. Social-emotional reciprocity	Abnormal social approach and failure of normal back-and-forth conversation; reduced sharing of interests, emotions, or affect; failure to initiate or respond to social interactions
	2. Nonverbal communicative behaviors used for social interaction	Poorly integrated verbal and nonverbal communication; abnormalities in eye contact and body language or deficits in understanding and use of gestures; total lack of facial expressions and nonverbal communication
	3. Developing, maintaining, and understanding relationships	Difficulties adjusting behavior to suit various social contexts; difficulties in sharing imaginative play or in making friends; absence of interest in peers
B. Restricted, repetitive patterns of behavior, interests, or activities, as manifested by at least 2 of the following, currently or by history; must have 2 of the 4 symptoms	1. Stereotyped or repetitive motor movements, use of objects, or speech	Simple motor stereotypies, lining up toys or flipping objects, echolalia, idiosyncratic phrases
	2. Insistence on sameness, inflexible adherence to routines, or ritualized patterns or verbal nonverbal behavior	Extreme distress at small changes, difficulties with transitions, rigid thinking patterns, greeting rituals, need to take same route or eat food every day
	3. Highly restricted, fixated interests that are abnormal in intensity or focus	Strong attachment to or preoccupation with unusual objects, excessively circumscribed or perseverative interest
	4. Hyper- or hyporeactivity to sensory input or unusual interests in sensory aspects of the environment	Apparent indifference to pain/temperature, adverse response to specific sounds or textures, excessive smelling or touching of objects, visual fascination with lights or movement

Symptoms must be present in the early developmental period (but may not become fully manifest until social demands exceed limited capacities or may be masked by learned strategies in later life). Symptoms cause clinically significant impairment in social, occupational, or other important areas of current functioning. These disturbances are not better explained by intellectual disability (intellectual developmental disorder) or global developmental delay. Intellectual disability and ASD frequently co-occur; to make comorbid diagnoses of ASD and intellectual disability, social communication should be below that expected for the general developmental level. Specify whether: with or without accompanying intellectual impairment, language impairment or associated with a known medical or genetic condition or environmental factor. Add code 293.89 if catatonia is also present. Reprinted with permission from the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (copyright 2013). American Psychiatric Association. All Rights Reserved.

Also with differening levels of severity: mild impairment vs severe impairment

Children - toddlers up to 5

Tips and Tricks for examining ears/ mouth-
DO IT LAST if possible

STABLIZE head and against cheek



Have parent help hold!

Figure 1. Examples of the oral examination of a child in the lap of a parent or guardian (top) and in the supine position (bottom).



Use tongue depressor: go in from the side, not front

Children - toddlers up to 5



Playtime

- Fine motor skills are improving, allow your toddler to pour things in one container and transfer to another. Also allow your child to scribble and draw.
- Hand-eye coordination improves and your child will have better direction and spatial awareness.
- Your child will participate in parallel play. Sharing will not start until later.
- Get down on your child's level and participate actively in playtime.
- Provide items for imaginative play, Ex. dress clothes, sand box, clay, water, crayons and music.

References

American Academy of Pediatrics. (n.d.). Feeding & Nutrition Tips: Your 2-Year-Old. HealthyChildren.org. Retrieved February 10, 2022, from <https://www.healthychildren.org/English/ages-stages/toddler/Pages/Feeding-and-Nutrition-Tips-For-Your-2-Year-Old.aspx>

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Farr, P. R. S. P. E., Farr, P. R. P. M. H. J., Farr, L. O. R. P. L., (Nag) R. D. C., Iuck, A. K. E. M. R. R., & Mun, C. M. R. B. C. (2017). *Maternal Child Nursing Care* (8th ed.). Mosby.

Gardner, Christine Vandenberg, Ruth E. Brune, Gill L. Crooks, Lesley, Durkin, Tanya, and R. Rasmussen. "Ages and Stages: Toddler (1-24 Months) (2104)". (2011) Faculty Publications from CHS, 74. <https://scholarcommons.usf.edu/childrens/74>

Toilet Training

- Start with toilet training and look for signs that they are ready and willing to learn. Signs are....
 - They stay dry for 2 hours
 - Do they know when they are wet or dry?
 - Can they pull pants up or down by themselves?
 - Can tell you and understand when they have to go. Encourage them to speak up!
- Schedule some potty breaks and plan for multiple trips. (up to 10x a day)
- Teach hand washing.
- Clean potty chairs every time your child uses them.
- Have your child choose underwear they are excited about.



ANTICIPATORY GUIDANCE

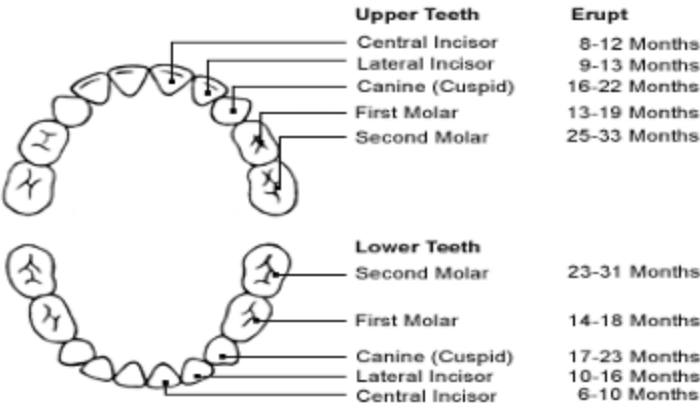
24 Months

1,200 x 927



Women • Infants • Children
WIC Program

Primary Teeth Eruption Chart



Thank you for watching!!

If any questions, please email me

Apate256@nyit.edu

Lecture Feedback:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



Resources:

1. Nelson Essentials of Pediatrics
2. The Harriett Lane Handbook Pediatrics
3. Netter's Pediatrics
4. Zitelli and Davis Atlas of Pediatric Physical Diagnosis
5. Pictures from variety of sources including UptoDate